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VIP **Homogeneous Catalysis** **Very Important Paper**

Photoredox/Cobalt-Catalyzed Chemo-, Regio-, Diastereo- and Enantioselective Reductive Coupling of 1,1-Disubstituted Allenes and Cyclobutenes

Qianghui Wu, Zhihan Zhang, Qinglei Chong, and Fanke Meng**

Abstract: A dual photoredox/cobalt-catalyzed protocol for chemo-, regio-, diastereo- and enantioselective reductive coupling of 1,1-disubstituted allenes and cyclobutenes through chemo-, regio-, diastereo- and enantioselective oxidative cyclization followed by stereoselective protonation promoted by a chiral phosphine–cobalt complex is presented. Such process represents an unprecedented reaction pathway for cobalt catalysis that enables selective transformation of the less sterically congested alkenes of 1,1-disubstituted allenes with cyclobutenes, incorporating a broad scope of tetrasubstituted alkenes into the cyclobutane scaffolds in up to 86% yield, >98:2 chemo- and regioselectivity, >98:2 dr and >99.5:0.5 er. Functionalization delivered a variety of enantioenriched cyclobutanes that are otherwise difficult to access. Preliminary mechanistic studies revealed that the reactions proceeded through oxidative cyclization followed by protonation and protonation might be the rate-determining step.

Enantioenriched cyclobutanes not only widely exist in natural products and pharmaceutical important molecules,[1] but also are useful intermediates and building blocks in organic synthesis (Scheme 1a).[2] Therefore, establishment of a general and modular strategy for catalytic enantioselective synthesis of a broad scope of densely functionalized cyclobutanes is highly sought after. Although catalytic enantioselective approaches for access to cyclobutanes include [2+2] cycloaddition,[3] cyclization,[4] multistep ring contraction and expansion reactions,[5] significant limitation on specific substrates patterns remained unaddressed. An alternative strategy is catalytic enantioselective functionalization of preformed four-membered carbocycles, comprising directed C–H functionalization as well as transformations of cyclobutanones and cyclobutenes.[6] The advantage for such strategy is that a high diversity of functional groups can be incorporated directly into the four-membered ring scaffold through a single set of transformations and starting materials.[7] Although numerous advances have been made in this field, significant limitations remained unsolved. A directing group was required for catalytic enantioselective C–H functionalization of cyclobutanes that enabled introduction of a boryl, aryl, or alkenyl group.[8] Only reduction and reactions of ketone enolates have been reported for catalytic enantioselective transformations of cyclobutanones.[9] Moreover, few approaches for catalytic enantioselective conjugate addition of polarized cyclobutenes, including Cu-catalyzed hydride, boryl, simple alkyl addition with dialkylzinc reagents,[10] organocatalyzed Diels–Alder reactions,[11] and Rh-catalyzed aryl addition with aryl boronic acids, [12] have been developed. We recently disclosed the first example of cobalt-catalyzed enantioselective alkyl conjugate addition of cyclobutenes with cobalt homoenolates generated from cyclopropanols.[13] A more challenging class of substrates are nonpolar cyclobutenes. The sluggish reactivity arose from relatively lower olefinic strain (1.9 kcal/mol) compared with that in cyclopropenes (27.7 kcal/mol) and bicyclic olefins (norbornene: 4.8 kcal/mol).[14] While metal-catalyzed enantioselective hydroboration,[15] borylallylation,[16] hydroamination,[17] and diboration[18] of cyclobutenes have been developed, enantioselective carbon–carbon bond forming reactions of nonpolar cyclobutenes remained scarce, and mainly limited to processes via carbometallation (Scheme 1b). Pioneering works Rh-catalyzed enantioselective arylation,[19] hydroacylation[20] of cyclobutenes as well as allylic substitution[21] of cyclobutanone ketals with aryl boronic acids have been reported by Fletcher and co-workers. Subsequently, we developed the first instance of cobalt-catalyzed enantioselective alkynylation and cascade allyl addition of cyclobutenes.[13] During the preparation of this manuscript, Lu and co-workers disclosed a Pd-catalyzed protocol for enantioselective arylation of cyclobutenes with aryl iodides.[22] However, protocols for catalytic enantioselective carbon–carbon bond forming

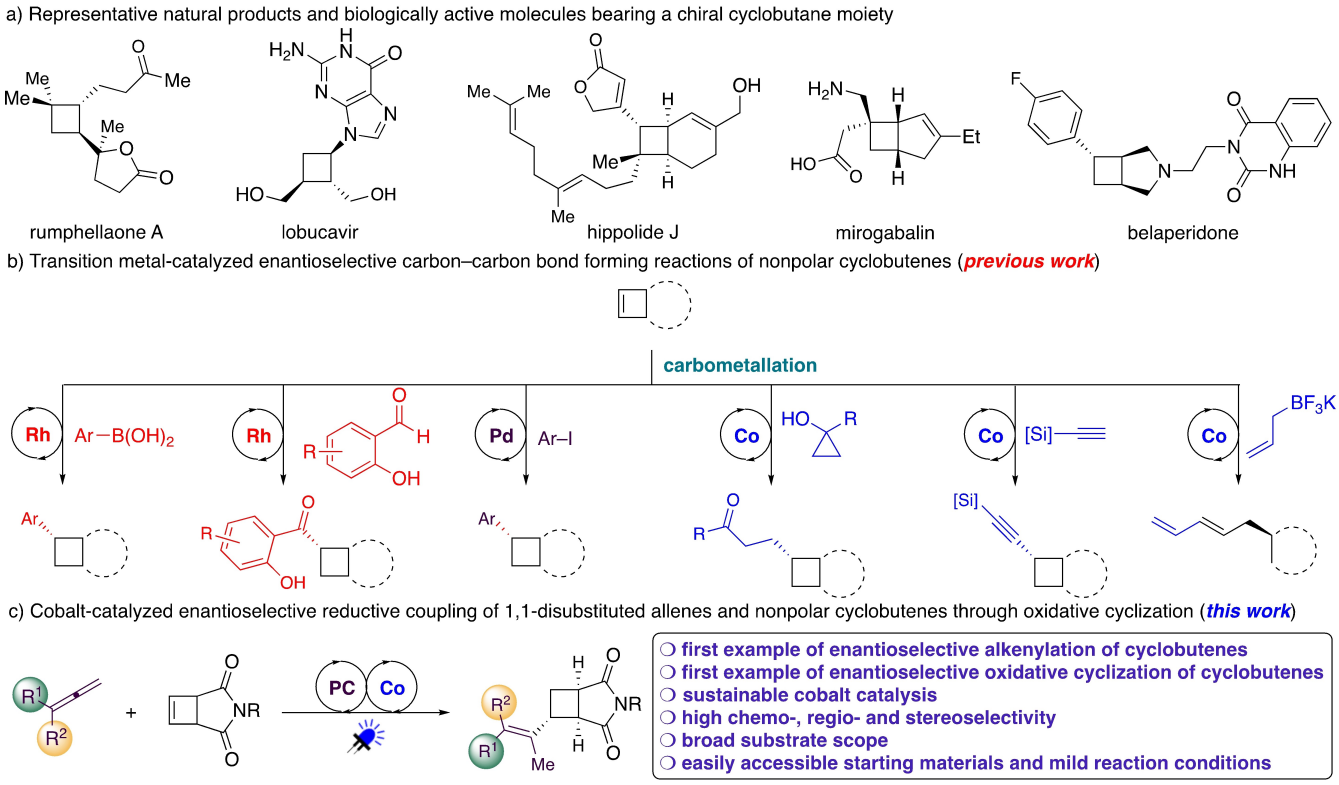
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Scheme 1. Reaction design

reactions of cyclobutenes through oxidative cyclization pathway remained elusive.

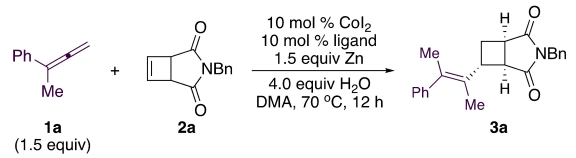
Oxidative cyclization of two π-bonds promoted by low-valent transition metal complexes represents a classical elementary step in organometallic chemistry. Transition metal-catalyzed transformations through oxidative cyclization constitutes one of the most important approaches for carbon–carbon bond forming reactions in organic synthesis, as such processes enable direct construction of carbon–carbon bonds from easily accessible unsaturated hydrocarbons without the requirement of pre-functionalization.[23] In this context, most researches focused on catalytic enantioselective reductive coupling of alkynes with aldehydes or activated alkenes through oxidative cyclization promoted by Ni-[24] and Co-based[25] catalysts. In contrast to reactions with alkynes that contain a single reactive site, it is rare and more challenging for accurate control of the chemo-, regio- and stereoselectivity of reactions with unsaturated hydrocarbons bearing multiple reactive sites. Although a few protocols for enantioselective coupling of 1,3-dienes through oxidative coupling have been reported,[26] transformations with allenes were far less developed. We disclosed the only example of cobalt-catalyzed regiodivergent enantioselective coupling of allenes and aldehydes through oxidative cyclization.[27] However, approaches for catalytic regio- and enantioselective reductive coupling of allenes with cyclobutenes remained elusive. One of the key challenges for such processes arises from the low reactivity of the nonpolar cyclobutenes. The inherent lower-lying LUMO of allenes results in preferential coordination to the metal center and higher reactivity for oxidative cyclization. Self-oligomerization of allenes has to be suppressed chemoselectively. Another crucial issue is accurate control of the regio- and enantioselectivity. We envisioned that a proper choice of chiral bisphosphine ligand combined with cobalt catalysis might be able to effectively tune the chemo-, regio- and enantioselectivity. Herein, we disclosed the first protocol for catalytic chemo-, regio- and enantioselective reductive coupling of 1,1-disubstituted allenes and cyclobutenes through oxidative cyclization, incorporating a variety of tetrasubstituted alkenyl groups and affording a broad scope of enantioenriched cyclobutanes that are otherwise difficult to access (Scheme 1c).

We commenced our studies with reaction of 1,1-disubstituted allene **1a** and cyclobutene **2a** in the presence of various Co complexes (0.2 mmol scale, see Supporting Information for more details) (Table 1). Zinc powder was added as the stoichiometric reductant. Axially stereogenic bisphosphine (**4a**), phosphinooxazoline (**4c**) and bisphosphine bearing a benzene backbone and chiral phospholane fragments (**4d**) were not able to promote the reaction (entries 1, 3–4), whereas reaction promoted by Co complex derived from chiral phosphine containing a ferrocene skeleton (**4b**) delivered the desired reductive coupling product **3a** in 21 % yield with 59:41 er (entry 2). Bisphosphine bearing an ethane backbone and chiral phospholane fragments (**4e**) provided high enantioselectivity albeit in low yield (entry 5). Introduction of stereogenicity into the diphenylphosphino ethane backbone instead of phospholane fragments (**4f–g**) resulted in 16–22 % yield with 51:49–73:27

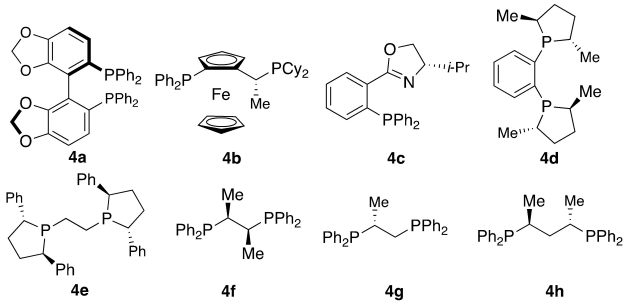
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Table 1: Optimization of reaction conditions.



entry	ligand	solvent	yield [%][a]	dr[b]	er[c]
1	**4a**	DMA	<5	NA[d]	NA[d]
2	**4b**	DMA	21	>98:2	59:41
3	**4c**	DMA	<5	NA[d]	NA[d]
4	**4d**	DMA	<5	NA[d]	NA[d]
5	**4e**	DMA	23	>98:2	1:99
6	**4f**	DMA	22	>98:2	49:51
7	**4g**	DMA	16	>98:2	27:73
8	**4h**	DMA	36	>98:2	98:2
9[e]	**4h**	DMA	47	>98:2	98:2
10[e]	**4h**	DMF	47	>98:2	99:1
11[e]	**4h**	NMP	59	>98:2	99:1
12[e]	**4h**	DMSO	62	>98:2	99:1
13[e]	**4h**	THF	46	>98:2	97:3
14[e]	**4h**	MeCN	46	>98:2	97:3
15[f]	**4h**	DMSO	69	>98:2	99:1
16[g]	**4h**	DMSO	82	>98:2	99:1
17[h]	**4h**	DMSO	80	>98:2	99:1
18[h][i]	**4h**	DMSO	<5	NA[d]	NA[d]
19[h][j]	**4h**	DMSO	<5	NA[d]	NA[d]
20[h][k]	**4h**	DMSO	19[l]	ND[m]	ND[m]
21[h][n]	**4h**	DMSO	<5	NA[d]	NA[d]
22[h][o]	**4h**	DMSO	<5	NA[d]	NA[d]
23[h][p]	**4h**	DMSO	45	>98:2	99:1



[a] Yield of isolated product. [b] Determined by analysis of ^{1}H NMR spectra of unpurified mixture. [c] Determined by analysis of HPLC spectra. [d] Not available. [e] 2.0 mol% 4CzIPN, 2.0 equiv. of *i*-Pr_2NEt, and 2.0 equiv. of Hantzsch ester irradiated by 40 W blue LED instead of Zn at 80 °C. [f] 2.0 mol% 4CzIPN, 2.0 equiv. of *i*-Pr_2NEt, and 2.0 equiv. of Hantzsch ester irradiated by 40 W blue LED instead of Zn at 40 °C. [g] 2.0 mol% 4CzIPN, 20 mol % of *i*-Pr_2NEt, 2.0 equiv. of Hantzsch ester and 0.5 equiv. of LiCl instead of Zn at 40 °C irradiated by 40 W blue LED. [h] 7.5 mol% CoI_2, 7.5 mol% **4h**, 2.0 mol% 4CzIPN, 15 mol% *i*-Pr_2NEt, and 2.0 equiv. of Hantzsch ester irradiated by 40 W blue LED instead of Zn at 40 °C. [i] Without light. [j] Without CoI_2. [k] Without **4h**. [l] Yield determined by analysis of ^{1}H NMR spectra of unpurified mixture with CH_2Br_2 as internal standard. [m] Not determined. [n] Without 4CzIPN. [o] Without Hantzsch ester. [p] Without *i*-Pr_2NEt.

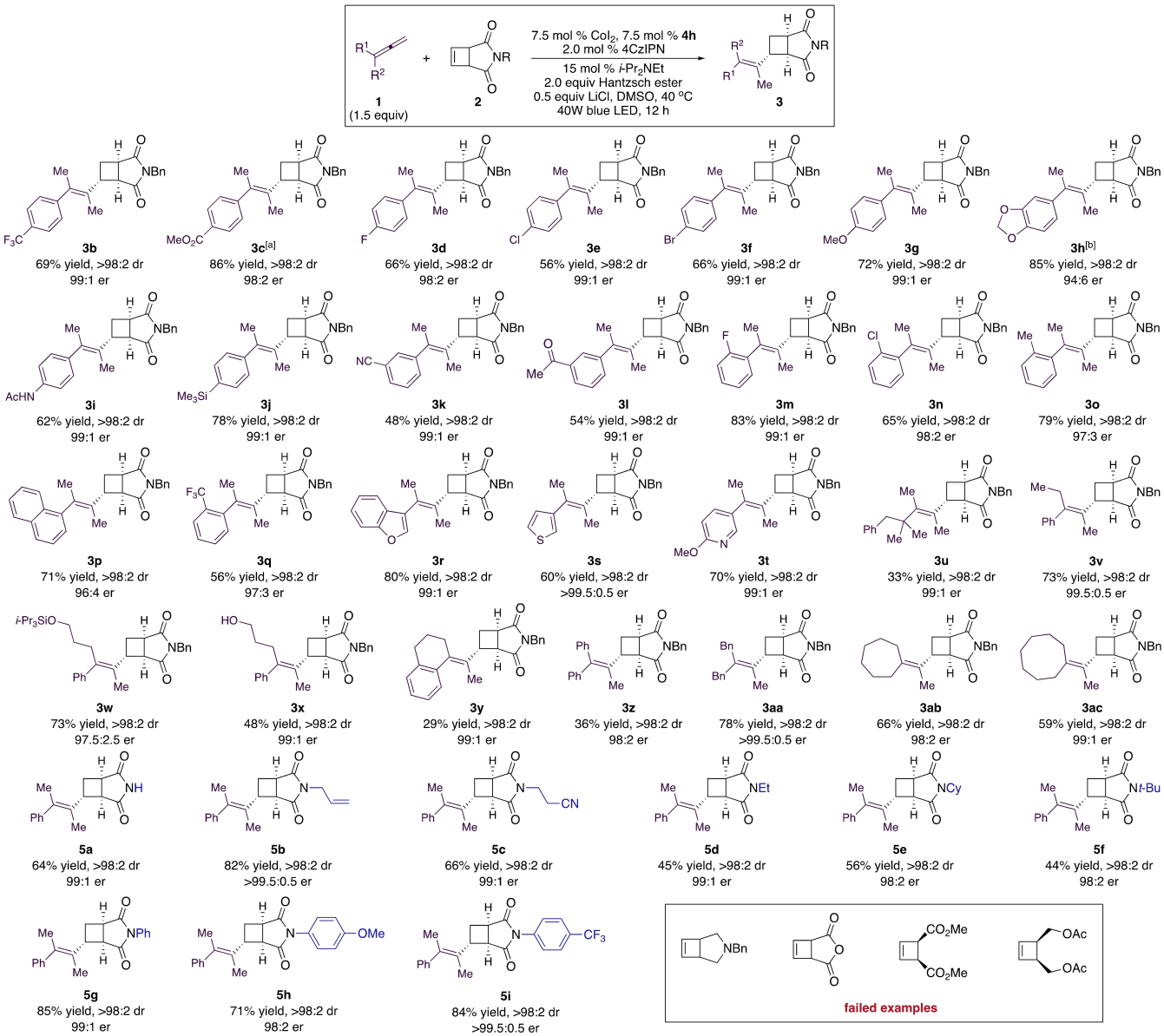
er (entries 6–7). Extending the length of the linker (**4h**) significant improvement for enantioselectivity (entry 8). Switching the reductant from zinc powder to photoredox conditions led to cleaner reaction and slight increase of efficiency (entry 9). Investigation on the solvent effect revealed that reaction performed in DMSO gave the highest yield without erosion of enantioselectivity (entries 10–14). Lowering the reaction temperature led to enhancement of efficiency (entry 15). Addition of LiCl resulted in further improvement of yield (entry 16). Reduction of catalyst loading wouldn't erode the efficiency (entry 17). It should be mentioned that a single (*E*)-isomer and diastereomer of product **3a** was obtained, and the major byproduct observed resulted from reduction of cyclobutene **2a**. Also, the less sterically congested alkene of the allene **1a** participated in the transformation chemo- and regioselectively. A series of control experiments were performed. In the absence of light, CoI_2, 4CzIPN, or Hantzsch ester, no reaction occurred (entries 18–19, 21–22). Without chiral ligand **4h**, 19% yield of **3a** was observed in the reaction mixture (entry 20). Removal of *i*-Pr_2NEt led to significant erosion of efficiency (entry 23).

With the optimal conditions in hand, we investigated the substrate scope (Scheme 2). 1,1-Disubstituted allenes containing electron-deficient (**3b–d**, **3k–m**, **3q**), electron-rich (**3g–i**), halogenated (**3d–f**, **3m–n**), and sterically congested (**3m–q**) aryl groups underwent chemo-, regio-, diastereo- and enantioselective reductive coupling, affording alkenyl-substituted cyclobutanes in 48–86% yield with 94:6–99:1 er.[28] Functional groups such as silyl (**3j**), ester (**3c**), ketone (**3l**) and cyano (**3k**) are well tolerated in the reaction conditions. 1,1-Disubstituted allenes bearing pharmaceutically important heterocycles (**3r–t**) served as suitable substrates. 1,1-Dialkyl-substituted allene bearing a tertiary alkyl was transformed into the desired product **3u** in 33% yield with 99:1 er. Reactions of 1,1-disubstituted allenes substituted with a larger alkyl group furnished cyclobutanes (**3v–y**) in 29–73% yield with 97.5:2.5–99.5:0.5 er as a single regio- and diastereomer. Allenes bearing two same substituents or carbocycles can participate in the transformation, producing the desired products (**3z**, **3aa–ac**) in 36–78% yield with 98:2–>99.5:0.5 er as a single regio- and diastereomer. It is worth mentioning that formation of (*Z*)-isomers of the products were observed in the reactions that produced **3c** and **3h**. For the other cases, (*E*)-isomers were obtained exclusively. Cyclobutenes containing succinimide without substitution (**5a**) and substituted with allyl (**5b**), functionalized alkyl (**5c**), primary alkyl (**5d**), secondary alkyl (**5e**) and tertiary alkyl (**5f**) as well as aryl (**5g–i**) groups are compatible with the reaction. It should be noted that similar with our previous report[27] and recent example developed by Lu and co-workers,[22] the fused succinimide moiety was required for the reactivity. No reaction occurred with cyclobutenes containing maleic anhydride.

The enantioenriched cyclobutanes that contain a tetrasubstituted alkene can be functionalized to furnish a variety of chiral building blocks that are otherwise difficult to access (Scheme 3). Although protocols for direct hydrogenation of the succinimide have been developed,[29] a multistep sequence was performed due to the possible intolerance of the alkene to hydrogenation conditions, affording diol **6** in 42% overall yield with 98:2 er (Scheme 3a).[30] The diol is versatile to be transformed into a variety of functional groups. Diastereoselective hydrogenation of the tetrasubstituted alkene promoted by Ir complex **7** provided cyclobutane **8** bearing a functionalized alkyl with two additional stereogenic centers in 91% yield with 90:10 dr

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Scheme 2. Substrate scope for cobalt-catalyzed chemo-, regio- and enantioselective reductive coupling of 1,1-disubstituted allenes and cyclobutenes. [a] A mixture of 89.5:10.5 (*E*/*Z*)-isomers were obtained. Yield of a mixture of (*E*/*Z*)-isomers. [b] A mixture of 89:11 (*E*/*Z*)-isomers were obtained. Yield of a mixture of (*E*/*Z*)-isomers.

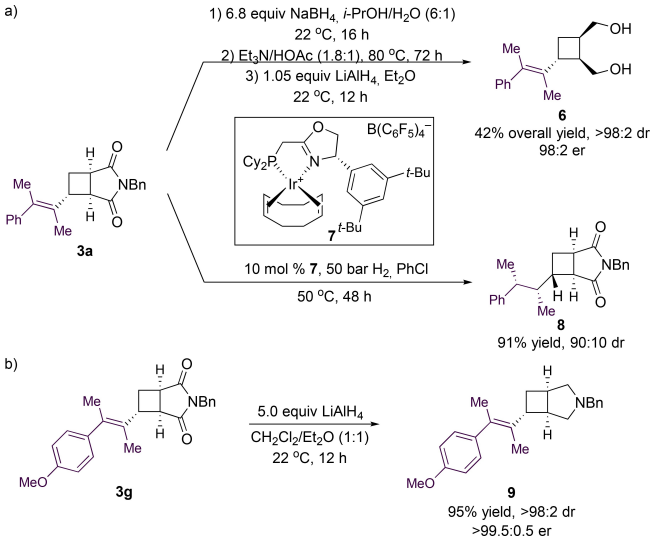
(Scheme 3a).[31] Reduction of the succinimide moiety with $LiAlH_4$ generated tetrahydropyrrole **9** without diminishment of stereoselectivity (Scheme 3b).

To gain some preliminary insights into the mechanism, we conducted a series of experiments (Scheme 4). Treatment of allene **1a** and cyclobutene **2a** with deuterated Hantzsch ester **10** afforded **11** in 40 % yield with 99:1 er, suggesting that direct protonation of the two C–Co bonds in the metallacycle intermediate occurred without possible β-hydride elimination (**IV**→**VI**, Scheme 5) and protonation of the stereogenic C–Co bond was stereoretentive (Scheme 4a). Reaction of deuterated cyclobutene **2a-D** provided **12** in 70 % yield with 99:1 er without scrambling deuterium, supporting that possible β-hydride elimination didn't happen (Scheme 4b). Transformation of deuterated cyclobutene **2a-D** in the presence of deuterated Hantzsch ester **10** delivered **13** in 50 % yield with 99:1 er (Scheme 4c). Overall, we were able to prepare a variety of deuterated products with different deuterium-incorporated sites simply by a proper choice of deuterated reagent and substrate. Parallel kinetic isotope effect experiments revealed that protonation of the C–Co bonds might be the rate-determining step (Scheme 4d). Competitive kinetic isotope effect experiments implied that protonation of the C–Co bond was irreversible (Scheme 4e).

To gain deepened understanding of reaction mechanism and probe the origin of selectivity control, density functional theory (DFT) calculations were performed for the oxidative cyclization process. As shown in Figure 1, given that propensity of first-row transition metals to exhibit multiple

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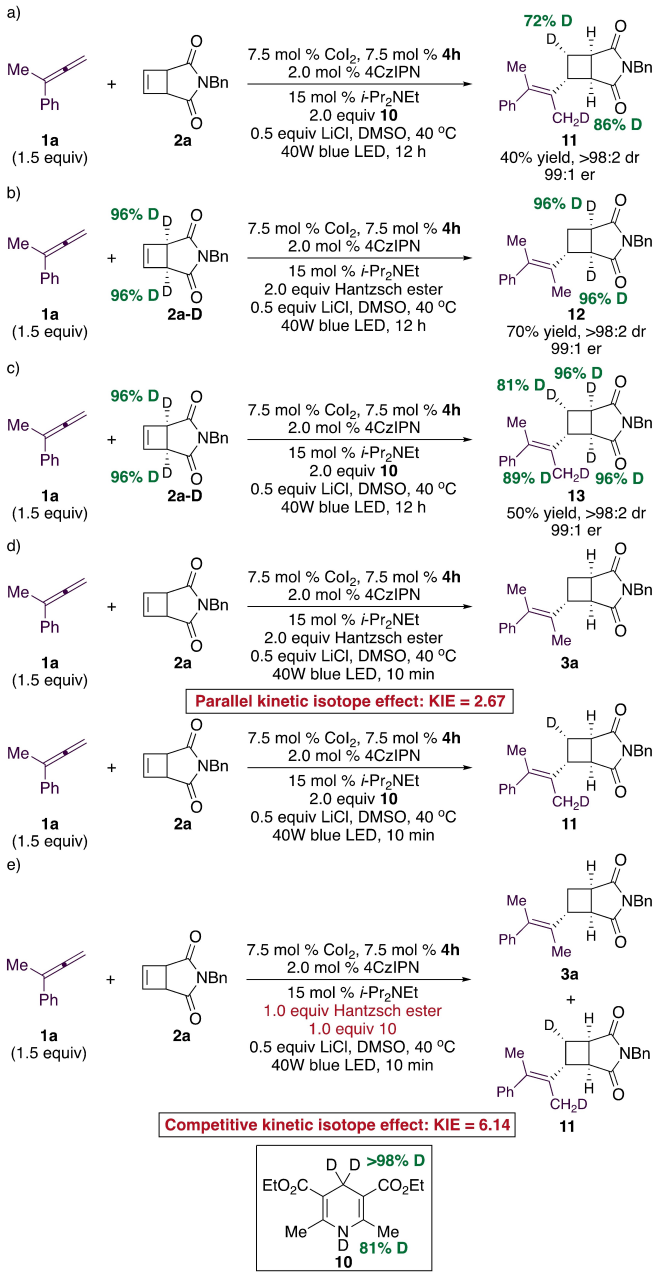
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Scheme 3. Functionalization

spin states close in energies, both singlet and triplet spin states were taken into accounts in our calculations. In both singlet and triplet Co(I) intermediates **INT-II**, allene and cyclobutene coordinates to metal center, while the triplet intermediate was found to be significantly more stable than its singlet counterpart. The triplet oxidative cyclization transition state ^{3}**TS-a** lies slightly lower in energy than singlet one ^{1}**TS-a**. The metallacycle intermediate ^{3}**INT-IV** was yielded after oxidative cyclization with driving forces of more than −10 kcal/mol (see Figure S1 for details of geometric structures in triplet). Here we focus on the oxidative cyclization transition states which represent stereo-determining process. The bottom of Figure 1 depicts the geometric structures for the most favorable transition state leading to product **3a** and its enantiomer. In the most favored configuration ^{3}**TS-a**, consistently reported by experiments and computations, the cyclobutene adopts an orientation towards to relative open site, mitigating close contacts with the upper left phenyl group of the ligand, allowing substrate fitting well with the chiral pocket of catalyst. Conversely, in the disfavored configuration ^{1}**TS-b**, where low spin state is more stable in this configuration, the cyclobutene points upward to the upper-left, leading to steric repulsion which forces a deviation of this phenyl group away from its optimum (see Figure S2 for Newman projection). Additional calculations for other plausible configurations of oxidative cyclization in both triplet and singlet states revealed they are higher in energy than ^{3}**TS-a**, agreeing well with the experimentally observed high regio- and stereoselectivities (see Figure S3). In different configurations, the relative stabilities of spin states were influences by the geometric arrangement between ligand and substrates.

Based on all the observations above, we proposed the catalytic cycles for the photoredox/cobalt-catalyzed chemo-, regio- and enantioselective reductive coupling of 1,1-disubstituted allenes and cyclobutenes (Scheme 5). Visible light irradiation excited 4CzIPN, which underwent single electron reduction by *i*-Pr$_2$NEt to afford 4CzIPN$^{\bullet-}$. 4CzIPN$^{\bullet-}$ was



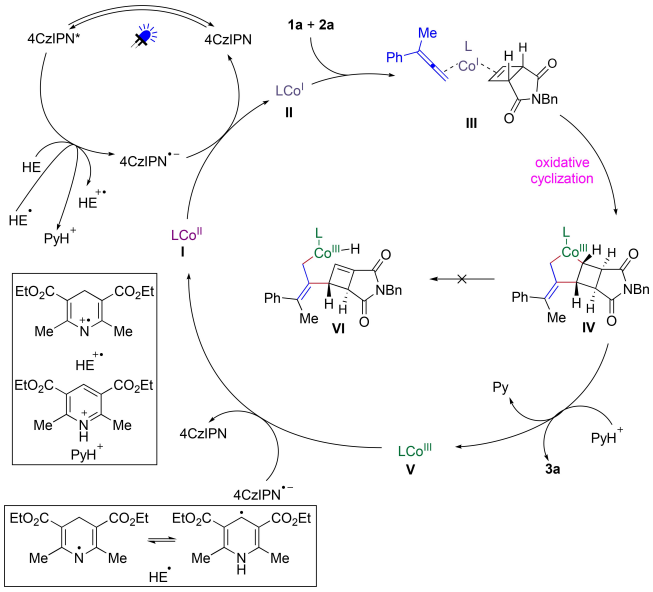
Scheme 4. Preliminary mechanistic studies

able to reduce the Co(II) complex **I** to Co(I) species **II**. Coordination of 1,1-disubstituted allene **1a** and cyclobutene **2a** to the Co center followed by oxidative cyclization delivered the metallacycle **IV**. Chemoselective protonation of the C–Co(III) bond provided the product **3a**. Possible β-hydride elimination of the metallacycle **IV** was not observed (**IV**→**VI**). Single electron reduction of the Co(III) complex **V** regenerated the Co(II) precursor **I**. It should be noted that another possible pathway that single electron reduction of the metallacycle **IV** to Co(II) intermediate followed by protonation of the C–Co(II) bonds cannot be ruled out.

In conclusion, we have developed a photoredox/cobalt-catalyzed protocol for chemo-, regio-, diastereo- and enantioselective reductive coupling of 1,1-disubstituted

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Scheme 5. Proposed catalytic cycle

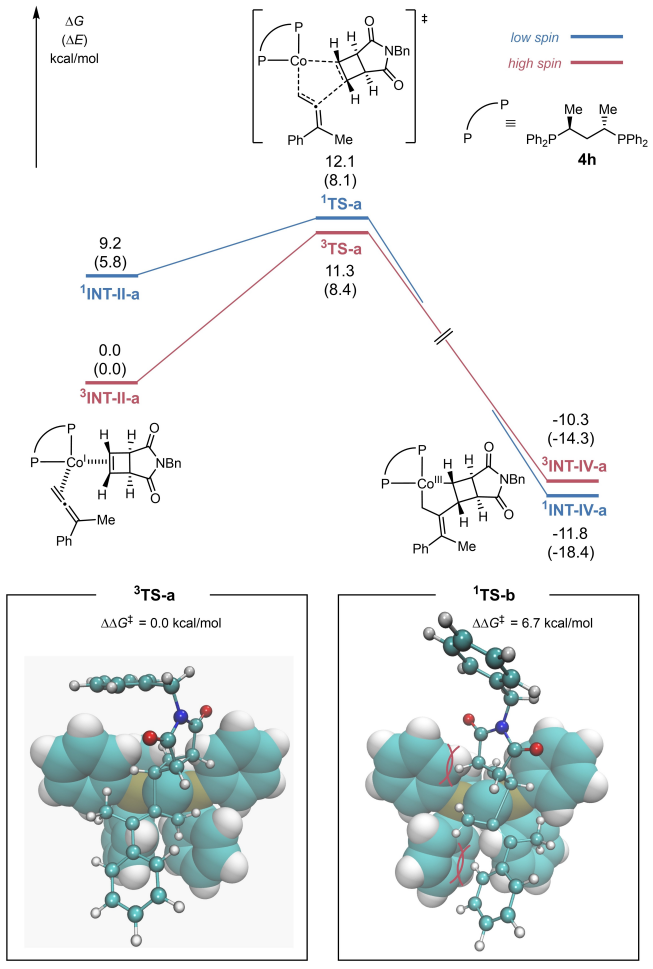


Figure 1. DFT calculations for oxidative cyclization step.

allenes and cyclobutenes to afford a broad scope of enantioenriched cyclobutanes in high efficiency, chemo-, regio- and stereoselectivity. To the best of our knowledge, it is the first time that reductive coupling of allenes and cyclobutenes through oxidative coupling pathway has been disclosed. In addition, such process represented the first example of generating hydroalkenylation products from enantioselective functionalization of cyclobutenes. Preliminary mechanistic studies revealed that the reaction proceeded through chemo-, regio- and stereoselective oxidative cyclization. Such discoveries unveiled a novel reaction pathway for cobalt catalysis, opening new possibilities for designing new transformations promoted by chiral cobalt-based catalysts. Further investigations on cobalt-catalyzed enantioselective reactions through oxidative cyclization are underway.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: allenes · cobalt · cyclobutenes · enantioselective catalysis · oxidative cyclization

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